Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

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Abstract

Many physical laws take the form "a known classical formula plus a small correction" (for example, relativistic time dilation). We present **ADCD** (Asymptotic Dictionary Correction Discovery), a method that searches for such a correction in noisy data. It differs from general-purpose symbolic regression in three ways: (i) each building block of the correction is written so that it is exactly zero in the classical limit, which avoids floating-point error; (ii) dimensional analysis proposes the dimensionless variables automatically; and (iii) a small table of physics-based function families restricts what is searched. ADCD then returns one of two verdicts: IDENTIFIABLE (the data support the discovered structure) or WITHHELD (they do not). On three synthetic tests with 1% noise, ADCD recovers the Screened Coulomb correction with an exact symbolic match ($\Delta\mathrm{BIC} = 30.65$; BIC = Bayesian Information Criterion), recovers the Entropy Expansion correction at the level of its function family ($\Delta\mathrm{BIC} = 14.70$; borderline, 2 of 5 noise seeds), and withholds judgment on Time Dilation at $v \leq 0.3c$, where measurement noise alone already accounts for a normalized mean squared error of $\mathrm{NMSE} \approx 0.41$, far above our 0.05 acceptance threshold. Results are reproducible given the same software and random seed.

Keywords: symbolic regression; correction discovery; dimensional analysis; structural prior; model selection; identifiability

1 Introduction

In the expanding landscape of machine learning for the physical sciences Carleo et al. [2019], Wang et al. [2023], physical laws frequently take a correction-first form: a known classical baseline extended by a term that vanishes at the classical limit Einstein [1905], Debye and Hückel [1923], Callen [1985]. Recovering that correction from observational data is distinct from full equation discovery: the baseline is fixed, only the residual must be identified, and that residual must vanish exactly where classical physics holds.

A concrete example. Coulomb's law gives an electrostatic force that falls off as $1/r^2$. In a simple phenomenological screening model, charges screen each other and the measured force falls off faster: $F = (k_e q_1 q_2 / r^2) e^{-r/\lambda_D}$. Given noisy measurements and the classical formula, ADCD returns the correction $\theta_0(e^{-r/\theta_4} - 1)$, where the fitted length $\theta_4 \approx 1.49$ m is close to the true Debye length (1.50 m). Just as important, when the data are too noisy to support any conclusion, ADCD says so instead of returning its best guess.

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General-purpose symbolic regression (SR) engines such as PySR Cranmer et al. [2020], Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], and PhySO Tenachi et al. [2023] excel at open-ended equation discovery from scratch. However, when repurposed directly to find small asymptotic corrections to an established law, unconstrained search encounters three practical hurdles: (i) *numerical cancellation*: near the classical limit $u \to 0$, standard representations suffer loss of floating-point significance IEEE Task P754 [2019]; (ii) *run-to-run variability*: stochastic optimization may return different Pareto candidates across random seeds Bongard and Lipson [2007], Schmidt and Lipson [2009]; and (iii) *uncalibrated confidence*: returning a best-fit expression without quantifying whether the residual signal exceeds the noise floor risks presenting an unverified expression as a physical law La Cava et al. [2021], Messeri and Crockett [2024], Krenn et al. [2022]. ADCD is designed specifically for this correction-first regime: algebraically regularized primitives eliminate numerical cancellation; deterministic grammar enumeration provides reproducible candidate evaluation; and explicit identifiability gating reports when data cannot discriminate between candidate structures.

Contributions.

- *Method.* A correction-first search whose building blocks vanish exactly in the classical limit, with dimensionless variables generated automatically.
- *Verdicts.* An explicit IDENTIFIABLE / WITHHELD decision based on four checks, so the method reports when the data are not informative enough.
- *Evidence.* Three synthetic scenarios (including one where ADCD correctly withholds), plus multi-seed and wrong-prior audits (Appendix A).

2 Related Work

Automated scientific discovery traces back to pioneering heuristic systems such as BACON Langley et al. [1987] and has evolved toward modern autonomous equation discovery Kramer et al. [2026], Makke and Chawla [2024], Karagiorgi et al. [2022]. General SR engines (Eureqa Schmidt and Lipson [2009], PySR Cranmer et al. [2020], Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], Udrescu et al. [2020], PhySO Tenachi et al. [2023]) target unconstrained equation recovery and typically do not enforce classical-limit constraints algebraically. SINDy Brunton et al. [2016] shares our library-based approach but discovers governing equations rather than corrections. PINNs Raissi et al. [2019], Karniadakis et al. [2021] embed physical laws as soft loss penalties; ADCD imposes them as hard algebraic identities before any numerical computation. Recent systems combining data with background knowledge and axiomatic reasoning, such as AI-Descartes Cornelio et al. [2023] and AI Hilbert Cory-Wright et al. [2024], highlight the importance of theory-grounded discovery, but, to our knowledge, do not address catastrophic cancellation near asymptotic limits or provide formal identifiability gating. While dimensional analysis and symmetry constraints have been integrated into SR and dynamical learning Tenachi et al. [2023], Petersen et al. [2021], Otto et al. [2025], ADCD adds a lightweight five-dimensional (MLTΘQ) dimensional grouping heuristic, transcendental argument safety, and BIC-based identifiability reporting in a single pre-fitting gate cascade. By combining deterministic grammar enumeration, algebraic classical-limit enforcement, phenomenon-specific functional priors, and explicit identifiability gating, ADCD provides a principled, falsifiable workflow tailored to the correction-first paradigm.

3 Methods

3.1 Problem formulation

Let y_{obs} and y_{cl} be the observed target and classical prediction. The correction residual is:

$$\Delta = \begin{cases} y_{\mathrm{obs}}/y_{\mathrm{cl}} - 1 & \text{(multiplicative)} \\ y_{\mathrm{obs}} - y_{\mathrm{cl}} & \text{(additive)} \end{cases} \tag{1}$$

To select between the two representations automatically, ADCD picks the form whose magnitude correlates less with the classical baseline, measured by the absolute Spearman rank correlation between $|y_{\mathrm{cl}}|$ and $|\Delta|$ (ties default to additive). There is no numeric ambiguity threshold: additive mode is used by construction when $\mathrm{std}(y_{\mathrm{cl}}) < 10^{-15}$ (a constant baseline), when fewer than 5 valid

points remain, or when both correlations vanish. ADCD searches for $\hat{\Delta}(\mathbf{x};\boldsymbol{\theta})$ such that $\hat{\Delta} \to 0$ as the relevant dimensionless ratio $u \to 0$.

3.2 Algebraically regularized primitive dictionary

Each primitive $D(u)$ satisfies $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity by construction, while rationalizing subtraction-prone forms against catastrophic cancellation. The reason is numerical: subtracting two nearly equal numbers wipes out significant digits, so a correction that should be tiny but nonzero can round to exactly zero. For example, the naive form $(1-u)^{-1/2} - 1$ collapses to 0.0 at $u = 10^{-16}$ in `float64`; the rationalized form $u/[\sqrt{1-u}(\sqrt{1-u}+1)]$ returns 5.0×10^{-17} at the same input. Table 1 lists all eight primitives.

Table 1: The eight algebraically regularized primitives, each satisfying $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity (for $\theta \neq 0$ in `D_pow`). The first five are the core physics-motivated basis; the remaining three extend coverage to oscillatory, saturation, and power-law regimes. Column "Physical origin" cites the fundamental equation that produces this functional family; it does not imply the primitive is restricted to that domain.

Name	Regularized form $D(u)$	Domain	Physical origin
`D_lor`	$u/[\sqrt{1-u}(\sqrt{1-u}+1)]$	$u \in [0, 1)$	Lorentz factor ($\gamma - 1$)
`D_rat`	$u/(1-u)$	$u \neq 1$	Yukawa/Fermi pole
`D_exp`	$e^{-u} - 1$	all u	Debye/Yukawa screening
`D_log`	$\ln(1+u)$	$u > -1$	Entropy of mixing
`D_sqrt_inv`	$1/\sqrt{1+u} - 1$	$u > -1$	Gravitational / Coulomb potential
`D_pow`	$\lvert u\rvert^{\lvert\theta\rvert}$	$u \geq 0$	Power-law corrections
`D_osc`	$1 - \cos u$	all u	Wave / AC oscillatory
`D_sat`	$\tanh u$	all u	Magnetic / sigmoid saturation

3.3 Dimensional grouping and free-scale generation

ADCD tracks the physical dimensions of every variable as a vector of integer exponents over mass (M), length (L), time (T), temperature (Θ), and electric charge (Q) (note: uppercase Θ denotes temperature dimension, distinct from lowercase model parameters θ_k). Instead of solving the full dimensional-analysis problem, it uses a simple two-step heuristic: first, it groups variables with identical dimensions into pairwise ratios (e.g., q_1/q_2); second, for each dimensional variable x_k (k=0, 1, ...), it generates two single-parameter free-scale ratios—linear x_k/θ_{2k} and quadratic x_k^2/θ_{2k+1}. Charge is tracked separately so that two charges can form a dimensionless ratio; a three-dimensional (MLT) engine would treat them as dimensionless scalars and skip this step. In Screened Coulomb, q_1 and q_2 consume θ_0–θ_3, so the distance ratios receive θ_4 and θ_5 (r/θ_4, r^2/θ_5). The index only reflects enumeration order; what matters is that the fitted value lands near the true Debye length.

3.4 Phenomenon-specific taxonomy as structural prior

Each physics phenomenon maps to a named set of permitted primitive families derived exclusively from its fundamental equations. For example, `yukawa_debye_screening` permits only `D_exp` and `D_rat`: the Klein-Gordon equation yields $V \propto e^{-mr}/r$ and Debye-Hückel theory yields $\phi \propto e^{-r/\lambda_D}/r$; both involve only exponential and rational families (applied phenomenologically to the Coulomb force in our benchmark). The taxonomy restricts *which family* is searched; it says nothing about *which variable* enters the argument or *what the scale is*—both are determined by the dimensional grouping heuristic and numerical optimizer. Allowing a function family that the physics does not justify gives the search an extra flexible function that can soak up noise, which shrinks the ablation ΔBIC gap and weakens identifiability. Table 2 lists entries for our three experiments.

A 3×3 cross-misspecification audit (Appendix A.3, Table 9) tests what happens when the wrong taxonomy is supplied. In all 6 wrong-taxonomy trials the system returned WITHHELD, so there were no false IDENTIFIABLE verdicts (the two Time Dilation trials fail the positive control under any prior; e.g., applying the oscillatory `wave_resonance` prior to Screened Coulomb yields WITHHELD with $\Delta\text{BIC} = -910.5$). In these tests, a wrong prior led to abstention rather than an invented law.

Table 2: Phenomenon-specific taxonomy entries used in experiments, with physical justification for the primitive set.

Phenomenon key	Primitives	Justification
`yukawa_debye_screening`	`D_exp`, `D_rat`	Klein-Gordon / Debye-Hückel: fundamental form is e^{-mr}/r.
`lorentz_special_relativity`	`D_lor`	Lorentz invariance of Maxwell equations uniquely selects the $\gamma - 1$ form.
`boltzmann_thermodynamics`	`D_exp`, `D_log`	Boltzmann partition function and entropy of mixing involve $e^{-\beta\epsilon}$ and $\ln$.

3.5 Grammar enumeration and theta-intact constraint

Candidates take the form $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$, optionally extended with a secondary additive or multiplicative-coupled term ($\theta_0 D_a(u) + \theta_1 D_b(u)$ or $\theta_0 D_a(u)[1 + \theta_1 D_b(u)]$; depth ≤ 3, ≤ 25 tokens). Dimensionless arguments are auto-generated by the dimensional grouping heuristic (up to 12 candidate ratios per scenario). Gates 1–3 (Table 3) inspect the symbolic *theta-intact* expression (where parameters θ_k remain symbolic without premature numerical substitution), treating single-variable free-scale ratios (e.g., r/θ_4 representing a Debye length) as dimensionally admissible because θ_k absorbs the variable's physical scale during fitting. Gate 4 (Asymptotic Regularization Check, ARC) evaluates the symbolic classical limit at $\boldsymbol{\theta} = \mathbf{1}$ (deferring parametric candidates for strict post-fit re-verification at $\hat{\boldsymbol{\theta}}$), and Gate 5 screens coarse numerical fitness at $\boldsymbol{\theta} = \mathbf{1}$.

3.6 Physical gate cascade

Before fitting, each candidate must pass Table 3: Gates 1–3 inspect the symbolic theta-intact expression; Gates 4–5 probe at $\boldsymbol{\theta} = \mathbf{1}$ (Gate 4 is re-verified post-fit at $\hat{\boldsymbol{\theta}}$).

Table 3: The five-gate pre-fitting physical cascade.

Gate	Rejection criterion
1. AST complexity	Post-expansion SymPy AST depth > 12 or node count > 50 (beyond grammar budget).
2. Dimensional consistency	Physical dimensions mismatch target (allowing free-scale x/θ_k).
3. Transcendental safety	Arguments to $\exp$, $\ln$, $\sqrt{\cdot}$ not dimensionless or θ_k-scaled.
4. Asymptotic check (ARC)	Classical limit at $\boldsymbol{\theta} = \mathbf{1}$ (and post-fit $\hat{\boldsymbol{\theta}}$) does not vanish.
5. Coarse numerical fitness	Evaluation at $\boldsymbol{\theta} = \mathbf{1}$ produces `NaN/Inf`.

3.7 Parameter optimization and model selection

The top-ranked Stage-1 surviving candidates ($K = 30$, shortlisted by coarse pre-fitting evaluation) are optimized via L-BFGS-B (JAX, `float64`). Parameters are log-reparameterized ($\theta_i = s_i e^{u_i}$, $s_i \in \{-1, +1\}$) for order-of-magnitude traversal; multiple restarts reduce sensitivity to local optima. Model selection uses an Extended BIC Schwarz [1978], Chen and Chen [2008]: $\mathrm{BIC} = k \ln N - 2 \ln \hat{L} + 2 \ln |\mathcal{C}|$ (implemented as $-2 \ln \hat{L} = N \ln \mathrm{NMSE}$, floored at 10^{-6}, up to a data-variance constant that cancels in $\Delta\mathrm{BIC}$), incorporating a multiple-testing penalty over the search space size $|\mathcal{C}|$. By the Kass–Raftery scale Kass and Raftery [1995], $\Delta\mathrm{BIC} > 10$ is the adopted identifiability threshold.

3.8 Four-step validation protocol

In plain terms. IDENTIFIABLE means the data are informative enough to single out the discovered function family. WITHHELD means they are not, so ADCD makes no structural claim even if it found a good-looking fit. Here, *domain-guided* refers to searching within the taxonomy prior, *ablated* searches all remaining primitives with the target family removed, and *Rank-1* denotes the candidate with the lowest Extended BIC. An IDENTIFIABLE verdict requires all four checks to pass:

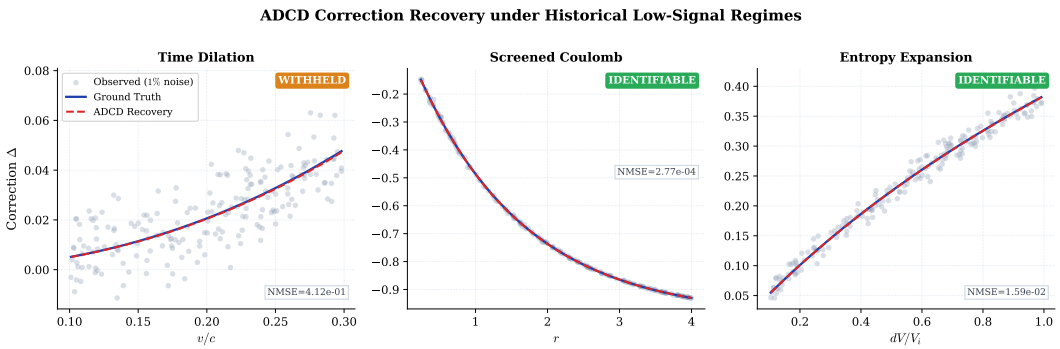


Figure 1: Correction recovery for the three scenarios. *Gray dots*: observed residual Δ at 1% Gaussian noise (relative residual $\Delta S/S_i$ for Entropy Expansion). *Blue solid*: ground truth. *Red dashed* ("ADCD Recovery"): the exact symbolic expression discovered by ADCD, evaluated with fitted parameters $\boldsymbol{\theta}$ (normalized by S_i for Entropy Expansion to match the relative scale); see Table 5 for exact NMSE values. Screened Coulomb (IDENTIFIABLE): exact symbolic match, NMSE $= 2.77 \times 10^{-4}$. Entropy Expansion (IDENTIFIABLE): class-level match, NMSE $= 1.59 \times 10^{-2}$. Time Dilation (WITHHELD): Lorentz structure found at Rank 1, NMSE $= 0.41$; positive control fails within the historical $v \leq 0.3c$ observation window.

1. **Budget disclosure.** Before looking at results, we state how many candidate formulas are searched ($|\mathcal{C}|$) and which function families are allowed, so the size of the search is visible.
2. **Positive control** (can the method recover the right answer at all?). Using only the true function family, the fit must explain at least 95% of the variance in the residual data (NMSE $=$ MSE/Var$(\Delta_{\text{obs}}) < 0.05$). If even the correct family fails, the data are too noisy to support a claim.
3. **Ablation control** (is the family really needed?). We remove the target family and search over all remaining primitives. We require the domain-guided candidate to beat the best remaining model by $\Delta\text{BIC} = \text{BIC}_{\text{ablated}} - \text{BIC}_{\text{domain}} > 10$ (conventionally "very strong evidence", Kass–Raftery). Extended BIC balances deviance improvement against parameter parsimony, penalizing superfluous complexity when competing functional forms achieve comparable fit (Table 7).
4. **Determinism.** Three runs with identical inputs must give identical output.

Failure of any check yields WITHHELD (in benchmark audits, Steps 2–3 evaluate against the known synthetic target family; in empirical deployment, Step 3 ablates the discovered candidate's family).

4 Experiments

4.1 Setup

Table 4 summarizes the three scenarios. All use 1% Gaussian multiplicative noise, $N = 200$ points, seed 42. Observation windows reflect historical low-signal regimes. The ground truth is never supplied to the blind discovery search (only noisy observations, classical predictions, dimensional definitions, and the taxonomy key are provided; the target family is referenced only in benchmark controls).

Table 4: Experimental scenarios with domains, historical observation windows, taxonomy keys, targets, and the full domain-guided search space size ($|\mathcal{C}|$).

Scenario	Domain	Window	Taxonomy key	Target	$\lvert\mathcal{C}\rvert$
Time Dilation	Relativity	$v \leq 0.3c$	`lorentz_special_relativity`	`D_lor`	10
Screened Coulomb	Electrostatics	$r \leq 4.0$ m	`yukawa_debye_screening`	`D_exp`	60
Entropy Expansion	Thermodynamics	$dV/V_i \leq 1.0$	`boltzmann_thermodynamics`	`D_log`	40

4.2 Results

Table 5 summarizes the quantitative results. Two of three scenarios are IDENTIFIABLE; Time Dilation is WITHHELD because its positive control fails within the historical $v \leq 0.3c$ observation window.

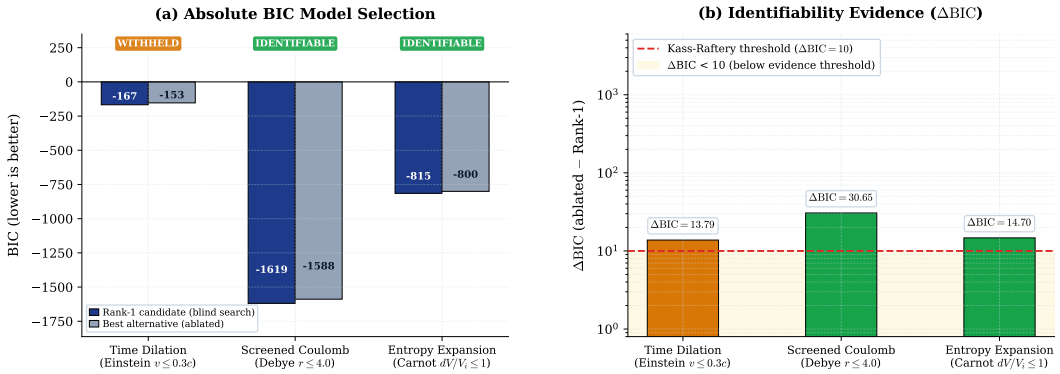


Figure 2: BIC identifiability comparison (seed $= 42$). *Left (a)*: absolute BIC of the Rank-1 domain-guided candidate (dark blue) vs. the best ablated alternative excluding the ground-truth family (gray). *Right (b)*: $\Delta\text{BIC} = \text{BIC}_{\text{ablated}} - \text{BIC}_{\text{best}}$ on a log scale; positive values favor the domain-guided Rank-1 candidate over the ablated best, with the Kass–Raftery "very strong evidence" threshold ($\Delta\text{BIC} = 10$, red dashed) Kass and Raftery [1995]. Although Time Dilation exceeds $\Delta\text{BIC} = 10$ via parsimony and search-space penalties ($\Delta D \approx 0$; Table 7), it is WITHHELD because its Positive Control fails (Table 5).

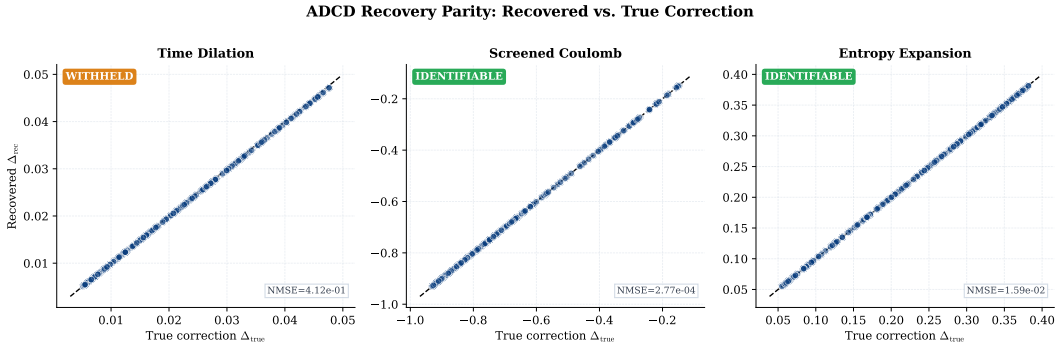


Figure 3: Parity plots: recovered correction (Δ_{rec}, evaluated with fitted parameters $\boldsymbol{\theta}$) vs. true correction (Δ_{true}) for $N = 200$ points (dashed: 1:1 line). All three scenarios track the 1:1 diagonal because domain-guided search placed the true structure at Rank-1 in every case. Crucially, Time Dilation achieves structural parity here yet is WITHHELD: its high error ($\text{NMSE} = 0.41$) on the *noisy* data (left panel of Fig. 1) fails the Positive Control, demonstrating principled abstention under insufficient SNR.

Reading ΔBIC. BIC scores a model by how well it fits the data, with penalties for extra parameters and for a larger search space. ΔBIC is the score of the best model *without* the true family minus that of the best model *with* it, so larger values favor the domain-guided result and values above 10 are conventionally called very strong evidence. It has three parts (Appendix Table 7): ΔD, the pure difference in fit; $\Delta k \ln N$, the reward for using fewer parameters; and a penalty for searching a larger space.[1] Reporting all three shows what drives each verdict.

Screened Coulomb. Recovered $\theta_0(e^{-r/\theta_4} - 1)$ is an exact symbolic match to the ground truth $\theta_0(e^{-r/\lambda_D} - 1)$, with $\theta_4 \approx 1.49$ m recovering the Debye length $\lambda_D = 1.50$ m. The engine selected the distance ratio r/θ_4 on its own (the index reflects enumeration order, as in Section 3.3). $\Delta\text{BIC} = 30.65$ far exceeds the threshold of 10, and $\Delta D = 20.14$ shows the advantage comes from fit quality, not from the search-space penalty. Over 5 noise seeds the verdict is IDENTIFIABLE every time (5/5); ΔBIC ranges from 11.0 to 263, and the mean of 76.61 is driven by one large value (penalty-free $\overline{\Delta D + \Delta k \ln N} = 71.39$).

[1] ΔBIC includes a search-space penalty $2 \ln |\mathcal{C}|$ that differs between domain-guided and ablated runs; ΔD reports the penalty-free fit-quality difference. Here $|\mathcal{C}_{\text{ablated}}| = 814/534/700$ for Screened Coulomb, Entropy Expansion, and Time Dilation, versus $|\mathcal{C}_{\text{domain}}| = 60/40/10$. Screened Coulomb's $\Delta D = 20.14$ alone exceeds the Kass–Raftery threshold of 10. Entropy Expansion's identifiability is parsimony-driven ($\Delta D = -1.08$; the ablated model achieves an equivalent fit with more parameters), as reported in Table 5.

Table 5: Four-step validation results for all three scenarios (all rows report Rank-1 candidates). PC = Positive Control (pass/fail). $\Delta\text{BIC} = \text{BIC}_{\text{ablated}} - \text{BIC}_{\text{best}}$, where BIC_{best} is the Rank-1 domain-guided search BIC; positive values indicate the domain-guided set is preferred over the ablated set. ΔD: penalty-free fit-quality difference (see footnote). Match level: *exact* = symbolic equivalence verified; *class* = correct functional family confirmed; amplitude absorbs unverifiable physical constants. Seed stability over 5 independent noise realizations: Screened Coulomb 5/5 IDENTIFIABLE; Entropy Expansion 2/5 IDENTIFIABLE (borderline at the Kass–Raftery boundary); Time Dilation 5/5 WITHHELD.

Scenario	Discovered structure	NMSE	PC	ΔD	ΔBIC	Match	Verdict
Screened Coulomb ($r \leq 4$ m)	$\theta_0(e^{-r/\theta_4} - 1)$	2.77×10^{-4}	PASS	20.14	30.65	exact	**IDENTIFIABLE**
Entropy Expansion ($dV/V_i \leq 1$)	$\theta_0 \ln(1 + dV/V_i)$	1.59×10^{-2}	PASS	-1.08	14.70	class	**IDENTIFIABLE** (2/5 seeds)
Time Dilation ($v \leq 0.3c$)	$\theta_0 \frac{v^2/c^2}{\sqrt{1-v^2/c^2}(1+\sqrt{1-v^2/c^2})}$	0.41	FAIL	0.00	13.79	class	**WITHHELD**

Entropy Expansion. Because the classical baseline S_i is constant, the scale-dependence test is uninformative and mode detection falls back to an additive search. Thus, the recovered additive correction $\theta_0 \ln(1+dV/V_i)$ matches the additive ground truth $nR\ln(1+dV/V_i)$ at class level: the fitted amplitude $\hat{\theta}_0 \approx 8.30$ J/K absorbs the composite constant $nR = 8.314$ J/K (within 0.2%). ADCD labels this match `class_only`: the functional family $\ln(1+u)$ is correctly identified, but because n and R are not individually supplied as variables to the search, the composite product nR cannot be decomposed into separate physical factors. $\Delta\text{BIC} = 14.70 > 10$ supports IDENTIFIABLE, but $\Delta D = -1.08$ shows the evidence comes from *simplicity*, not from a better fit: the ablated search matches the fit with k=3 parameters, versus k=1 for the domain-guided model. Over 5 noise seeds the verdict is borderline (2/5 IDENTIFIABLE, $\overline{\Delta\text{BIC}} = 9.91 \pm 3.40$, sample SD), so the logarithmic correction sits at the decision boundary under 1% noise.

Time Dilation. Over $v/c \in [0.10, 0.30]$, the Lorentz correction $\gamma - 1$ peaks at 4.8% with a signal standard deviation of $\text{SD}(\Delta_{\text{true}}) \approx 1.25\%$ against the 1.0% measurement noise. The grammar recovers the Lorentz structure $D_{\text{lor}}(v^2/c^2)$ at Rank 1 ($\hat{\theta}_0 \approx 1.01$), but the positive control fails ($\text{NMSE} = 0.41$). Put simply, even the exact true curve would differ from the noisy data by about 41% of the data's variance ($\text{NMSE} \approx 0.412 > 0.05$), so no method could pass the 5% test in this window; the Positive Control correctly returns WITHHELD. Over 5 noise seeds this verdict is stable (5/5 WITHHELD).

5 Discussion

Why withholding is a useful answer. A method that always returns a formula, whatever the data quality, cannot tell the reader when to trust it. ADCD's WITHHELD verdict on Time Dilation marks the limit of the data: lowering noise ($\sigma \lesssim 0.28\%$) or widening the velocity window ($v \gtrsim 0.52c$ at $\sigma = 1\%$) reduces the noise floor below 0.05, while separating D_{lor} from competing rational/polynomial families on pure fit ($\Delta D > 10$) is expected to require extending into the relativistic regime ($v > 0.3c$).

Taxonomy as prior, not hint. The taxonomy restricts *which primitive family* is active; the dimensional grouping heuristic and optimizer determine *which variable* and *which scale* within that family. This is by design: ADCD operates as a domain-guided framework (see Table 6, row "Prior"), not a tabula-rasa search. The taxonomy prior is active from the budget-disclosure step onward, consistent with the framework's explicit use of peer-reviewed functional priors as a structural constraint. In Screened Coulomb, the search, not the taxonomy, selected the distance variable and its scale ($\theta_4 \approx 1.49$ m). In the 3×3 misspecification audit, none of the 6 wrong-taxonomy trials produced a false IDENTIFIABLE verdict.

Comparison with general SR. ADCD and general SR address complementary objectives: discovering small corrections to an established baseline vs. discovering full governing equations from scratch. Unconstrained SR explores broad hypothesis spaces and constructs a Pareto frontier of complexity versus accuracy. In contrast, ADCD formally tests whether observational data warrant a specific structural correction to a mature classical baseline. Table 6 summarises these complementary operational regimes.

Principle-guided discovery. Recent perspectives on AI for physics stress that theory building needs compact laws, symmetry, and controlled limits Shalyt et al. [2026]. ADCD follows this spirit: it

searches for a small correction anchored to an established baseline and reports how strongly the data support it.

Table 6: Complementary roles of ADCD and general-purpose symbolic regression.

Property	ADCD	General SR
Target	Corrections to known laws	Full equations from scratch
Prior	Phenomenon-specific taxonomy	None (tabula rasa)
Dimensional engine	5D MLTΘQ	Varies (none to full unit constraints, e.g., PhySO)
Classical limit	Algebraic guarantee	Soft loss penalty
Output	Ranked top-K + verdict	Pareto frontier
Identifiability	Explicit (WITHHELD if insufficient)	None built in
Reproducibility	Deterministic (fixed seed/env)	Stochastic (seedable)

5.1 Limitations

Weak-signal convergence. Time Dilation's positive control failure is the intended output when the signal-to-noise ratio is too low. Future work should investigate adaptive restarts and denser initialization grids.

Scope. Current support covers eight primitive families, additive and multiplicative corrections, and synthetic Gaussian noise. Heavier-tailed noise and real observational data (e.g., galactic rotation curves, pulsar timing residuals) remain future work; real data require preprocessing for heteroscedastic uncertainties not yet modelled. Validation thresholds ($\Delta\mathrm{BIC} > 10$, $\mathrm{NMSE} < 0.05$) were evaluated under $N = 200$ and 1% Gaussian noise; calibrating across broader noise regimes and sample sizes remains future work.

Taxonomy coverage. Each new entry requires a peer-reviewed derivation for every admitted family—a deliberate constraint preventing unchecked hypothesis-space expansion.

Safety–expressivity trade-off. ADCD constrains its hypothesis space relative to general-purpose SR: candidates failing dimensional consistency or outside the phenomenon-specific primitive set are rejected before fitting. This prevents fitting spurious mathematical complexity to observational noise, but means ADCD cannot discover corrections whose functional family lacks a documented derivation. The appropriate framing is complementarity: general SR explores broad hypothesis spaces from scratch; ADCD audits targeted corrections to an already-established baseline.

Seed stability of borderline verdicts. Entropy Expansion's verdict varies across noise realizations (2/5 IDENTIFIABLE, 3/5 WITHHELD), indicating the logarithmic correction sits at the identifiability boundary under 1% noise. We treat this as a limitation of the current noise regime.

6 Conclusion

ADCD is a deterministic, correction-first framework combining algebraically regularized primitives, 5D (MLTΘQ) dimensional grouping, and a Phenomenon-Specific Taxonomy. On three canonical scenarios, two yield IDENTIFIABLE results (Screened Coulomb: $\Delta\mathrm{BIC} = 30.65$, exact match; Entropy Expansion: $\Delta\mathrm{BIC} = 14.70$, class-level, parsimony-driven; seed 42, 2/5 seeds IDENTIFIABLE) and one yields a principled WITHHELD verdict (Time Dilation at $v \leq 0.3c$), where the historical signal is insufficient for a confident structural claim. Within its deliberate scope—known classical baseline, additive or multiplicative correction, Gaussian noise—ADCD provides deterministic reproducibility and explicit identifiability diagnostics, returning a principled WITHHELD verdict (with its $\Delta\mathrm{BIC}$) when observational data cannot discriminate candidate structures. We hope this work encourages broader adoption of explicit identifiability checks and abstention mechanisms in scientific machine learning.

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Use of AI-Assisted Tools

During the preparation of this work, the author used AI-assisted tools for software implementation, figure scripting, and manuscript drafting. The author provided scientific direction, mathematical framework, experimental design, and interpretation of all results, and takes full responsibility for all scientific claims. No AI tool is listed as an author. The codebase and raw validation logs are publicly available at https://github.com/apiprdt/PhysicsPaper.

Data Availability

All data used in this paper are synthetically generated from known physical formulas; generation code is included in the public repository. No proprietary or restricted datasets are used.

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A Supplementary Empirical Audits

This appendix provides the complete numerical tables for the three empirical audits referenced in the main text: (1) the Extended BIC deviance and penalty decomposition (Section A.1), (2) the five-seed noise realization stability analysis (Section A.2), and (3) the 3×3 phenomenon-taxonomy cross-misspecification safety audit (Section A.3).

A.1 Extended BIC Deviance and Penalty Decomposition

Under the Extended BIC formulation ($\mathrm{BIC} = k \ln N + N \ln \mathrm{NMSE} + 2 \ln |\mathcal{C}|$), the ablation gap $\Delta\mathrm{BIC} = \mathrm{BIC}_{\mathrm{ablated}} - \mathrm{BIC}_{\mathrm{domain}}$ decomposes into pure fit quality ($\Delta D = N \ln(\mathrm{NMSE}_{\mathrm{ablated}}/\mathrm{NMSE}_{\mathrm{domain}})$), parameter parsimony ($\Delta k \ln N$), and search-space multiplicity ($2 \ln(|\mathcal{C}_{\mathrm{ablated}}|/|\mathcal{C}_{\mathrm{domain}}|)$), as reported in Table 7 for $N = 200$ ($\ln 200 \approx 5.30$, seed $= 42$).

Table 7: Exact decomposition of $\Delta\mathrm{BIC}$ into pure data-fit deviance (ΔD), parameter parsimony ($\Delta k \ln N$), and search-space multiplicity penalty ($2 \ln(|\mathcal{C}_{\mathrm{abl}}|/|\mathcal{C}_{\mathrm{dom}}|)$) at $N = 200$, seed $= 42$ (components may differ from totals by 0.01 due to rounding).

Scenario	$\lvert\mathcal{C}_{\mathrm{dom}}\rvert$	$\lvert\mathcal{C}_{\mathrm{abl}}\rvert$	$k_{\mathrm{dom}} \to k_{\mathrm{abl}}$	ΔPenalty	$\Delta k \ln N$	ΔD	ΔBIC	Primary Driver
Screened Coulomb	60	814	$2 \to 3$	$+5.22$	$+5.30$	$+\mathbf{20.14}$	30.65	Pure fit ($\Delta D > 10$)
Entropy Expansion	40	534	$1 \to 3$	$+5.18$	$+10.60$	-1.08	14.70	Parameter parsimony
Time Dilation	10	700	$1 \to 2$	$+8.50$	$+5.30$	0.00	13.79	Penalty + parsimony (WITHHELD)

A.2 Five-Seed Noise Realization Stability

To evaluate sensitivity to random measurement fluctuations under 1% Gaussian noise, the complete four-step validation protocol was executed across five independent noise seeds (seed $\in \{0, 42, 99, 256, 512\}$). Table 8 summarizes the per-seed $\Delta\mathrm{BIC}$ trajectory, mean and sample standard deviation (s, $\mathrm{ddof} = 1$), penalty-free fit advantage ($\overline{\Delta D + \Delta k \ln N}$), and verdict breakdown.

Table 8: Verdict and $\Delta\mathrm{BIC}$ stability across five independent 1% Gaussian noise realizations (seed $\in \{0, 42, 99, 256, 512\}$). Standard deviations denote unbiased sample standard deviations (s, $\mathrm{ddof} = 1$; Screened Coulomb: median $\Delta\mathrm{BIC} = 30.65$, while the mean is pulled upward by the seed-99 outlier).

Scenario	Seed 0	Seed 42	Seed 99	Seed 256	Seed 512	$\overline{\Delta\mathrm{BIC}} \pm s$	$\overline{\Delta D + \Delta k \ln N}$	Verdict Breakdown
Screened Coulomb	56.25	30.65	263.16	11.02	21.94	76.61 ± 105.62	$+71.39$	**5/5 IDENTIFIABLE**
Entropy Expansion	9.50	14.70	5.12	9.91	10.34	9.91 ± 3.40	$+4.73$	2/5 IDENT., 3/5 WITHHELD
Time Dilation	8.35	13.79	5.49	8.01	13.73	9.87 ± 3.72	$+1.38$	**5/5 WITHHELD**

A.3 Taxonomy Cross-Misspecification Safety Audit

To verify that the phenomenon-specific taxonomy prior does not induce false discoveries when misspecified, each scenario was evaluated under its true physical taxonomy key and two intentionally mismatched keys that exclude the ground-truth primitive family. As shown in Table 9, all six misspecified trials returned **WITHHELD** (0/6 false discoveries): when the true primitive family is absent from the domain prior, the unrestricted ablation search outperforms the misspecified domain search ($\Delta\mathrm{BIC} \in [-1038.24, -674.62]$ on identifiable scenarios), triggering conservative abstention rather than overfitting.

Table 9: 3×3 taxonomy cross-misspecification safety audit (seed $= 42$). PC Pass tests the scenario's target family ($\mathrm{NMSE}_{\mathrm{PC}} < 0.05$), while $\mathrm{NMSE}_{\mathrm{best}}$ and $\Delta\mathrm{BIC}$ reflect the blind search under the supplied prior; negative $\Delta\mathrm{BIC}$ values indicate that the unrestricted ablation set outperforms the misspecified prior, triggering WITHHELD.

Scenario	Prior Type	Supplied Taxonomy Key	Permitted Primitives	PC Pass	$\mathrm{NMSE}_{\mathrm{best}}$	ΔBIC	Verdict
Screened Coulomb	Correct	`yukawa_debye_screening`	`D_exp, D_rat`	PASS	2.77×10^{-4}	$+30.65$	**IDENTIFIABLE**
Screened Coulomb	Wrong A	`wave_resonance`	`D_osc, D_rat`	PASS	2.99×10^{-2}	-910.51	WITHHELD (Abl. fail)
Screened Coulomb	Wrong B	`critical_scaling`	`D_pow`	PASS	2.93×10^{-2}	-903.76	WITHHELD (Abl. fail)
Entropy Expansion	Correct	`boltzmann_thermodynamics`	`D_exp, D_log`	PASS	1.59×10^{-2}	$+14.70$	**IDENTIFIABLE**
Entropy Expansion	Wrong A	`lorentz_special_relativity`	`D_lor`	PASS	3.13×10^{0}	-1038.24	WITHHELD (Abl. fail)
Entropy Expansion	Wrong B	`wave_resonance`	`D_osc, D_rat`	PASS	4.87×10^{-1}	-674.62	WITHHELD (Abl. fail)
Time Dilation	Correct	`lorentz_special_relativity`	`D_lor`	FAIL	4.12×10^{-1}	$+13.79$	WITHHELD (PC fail)
Time Dilation	Wrong A	`boltzmann_thermodynamics`	`D_exp, D_log`	FAIL	4.13×10^{-1}	$+9.96$	WITHHELD (PC fail)
Time Dilation	Wrong B	`yukawa_debye_screening`	`D_exp, D_rat`	FAIL	4.12×10^{-1}	$+10.53$	WITHHELD (PC fail)